

WhatsApp AI Agent

Multi-Tenant SaaS Platform

Business Requirements Document

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This document defines the complete business requirements for the WhatsApp AI Agent platform — a multi-tenant SaaS product enabling businesses to deploy AI-powered WhatsApp bots and voice agents for customer support, sales engagement, and lifecycle management, with embedded payment collection and campaign tooling.

1. Executive Summary

WhatsApp AI Agent is a B2B SaaS platform that enables businesses to deploy AI-powered conversational bots on WhatsApp and Voice channels without writing any code. The platform provides three specialised bot personas — Customer Support, Sales Engagement, and Lifecycle Management — each configurable through a no-code dashboard with layered guardrails, custom knowledge bases, escalation policies, voice call pipelines, campaign tools, and embedded payment collection.

Built for the Indian SMB and mid-market, the platform integrates natively with local payment gateways (PhonePe, Razorpay), local telephony (Exotel), and India's regional language AI providers (Sarvam AI). It supports 10+ regional languages and deploys on top of the Meta WhatsApp Business API.

Core Value Propositions

- Zero-code bot deployment — businesses configure, not code.
- Multi-bot, multi-tenant — one platform serves unlimited client organisations.
- 4-layer guardrail cascade — platform → product → client → bot safety controls.
- Confidence-driven escalation — AI self-reports uncertainty and hands off to humans automatically.
- Omnichannel — WhatsApp messaging + outbound/inbound AI voice calls in one product.
- Embedded commerce — orders, UPI/card payment links, and delivery tracking inside conversations.
- India-first stack — Exotel, PhonePe, Razorpay, Sarvam AI, 10+ regional languages.
- Enterprise-grade security — Supabase Row-Level Security with per-tenant data isolation.

2. Platform Architecture

2.1 System Components

Component	Technology	Responsibilities
API Server (apps/api)	Node.js / Fastify	Webhook processing, AI response pipeline, voice orchestration, campaign dispatch, KB ingest.
Web Application (apps/web)	Next.js 14 App Router	Dual portal: client dashboard + platform admin UI. Server Actions for all data mutations.
Shared Package	TypeScript	Cross-package types, enums, and shared utility functions.
Database Package	Supabase / PostgreSQL	Migrations, RLS policies, pgvector for embeddings, get_bot_context RPC.
Cache Layer	Redis	KB lookup cache, bot context cache (60-second TTL), token quota counters.

2.2 User Roles & Permissions

Role	Scope	Permissions
Platform Manager	SaaS Operator	Full access: create/manage clients, products, guardrails, LLM configs, voice providers, billing, notifications, platform staff.
Platform Admin	SaaS Operator	Read-only monitoring view of all client data and system health.
Client Manager	Tenant	Full workspace control: bot configs, guardrails, KB, team, WhatsApp numbers, LLM keys, orders.
Client Admin	Tenant	Escalation management, limited team management within the tenant.
Agent	Tenant	View and claim assigned escalations; send replies during human takeover.

2.3 Tenant Plans

Plan	Conversation Limit / Month	AI Token Quota
Starter	500 conversations	Monthly cap enforced (plan-based)
Growth	2,000 conversations	Higher monthly cap (plan-based)
Scale	Unlimited	Unlimited

Tenant status: trial → active (on subscription) or trial → expired. Suspended tenants receive no bot replies; existing conversation history is preserved.

3. Bot Types (Products)

The platform ships three distinct bot products. Each can be activated independently per client with its own system prompt, knowledge base, guardrails, voice settings, LLM credentials, and escalation policy.

Customer Support Bot — slug: support_bot

Resolves customer queries by answering FAQs, looking up order/account status, and handling common support workflows via the knowledge base.

- FAQ resolution from the knowledge base
- Order and account status lookup
- Confidence-based escalation to human agents
- CSAT rating collection after resolution (1–5 stars)
- Multilingual support (10+ Indian regional languages)

Sales Engagement Bot — slug: sales_bot

Qualifies inbound leads, shares product information, and detects buying intent to hand off warm prospects to a sales agent at exactly the right moment.

- Lead qualification through structured conversation stages
- Product information delivery from the KB
- Sales lead detection ([SALES_LEAD] marker) with auto-escalation to sales agent
- Customer entity capture (email, product interest, quantity, location)
- Conversation state machine: greeting → qualifying → resolving → following_up → closing

Lifecycle Management Bot — slug: lifecycle_bot

Handles post-sale engagement: order tracking, invoice queries, payment reminders, and renewal outreach integrated with the Orders and Payments module.

- Order status and delivery tracking
- Invoice and payment query resolution
- Payment link dispatch (PhonePe / Razorpay) within the conversation
- Proactive follow-up reminders for pending payments
- Renewal and upsell outreach

4. Core Feature Modules

4.1 Inbound WhatsApp Message Pipeline

Every inbound WhatsApp message triggers the pipeline below. The HTTP 200 response is sent immediately; all AI processing is fully asynchronous.

- 1. Provider Inference** Payload shape determines the WhatsApp provider (Meta Cloud vs Twilio). Falls back to the alternate provider if no active number is found for the inferred one.
- 2. Bot Context Load** Single `get_bot_context` RPC fetches: WhatsApp number credentials, tenant plan/status, bot config, all 4 guardrail layers, and LLM credentials. Result Redis-cached for 60 s.
- 3. Plan & Status Enforcement** Suspended tenants silently dropped. Monthly conversation and token quotas checked

4. Contact Upsert	Contact identified by phone number (E.164) or BUID (opaque WhatsApp username ID). Contact record and persistent memory JSON are created or updated.
5. CSAT Intercept	If contact has an awaiting_csat flag and replies 1–5, the rating is recorded and a thank-you confirmation is sent. No AI call is made.
6. Message Deduplication	whatsapp_msg_id unique constraint prevents duplicate AI calls when Meta retries webhooks.
7. Optimistic Processing Lock	30-second lock on the conversation row prevents concurrent AI calls from simultaneous retries.
8. Escalation Keyword Check	Configured trigger phrases checked before AI call. Match → conversation escalated, human-handover message sent immediately.
9. KB Retrieval (RAG)	4-strategy cascade: Redis cache → pgvector similarity search (Voyage-3 embeddings) → keyword ILIKE search → product-scoped legacy fallback.
10. History Assembly	Up to 40 messages loaded. Recent messages passed verbatim; older turns compressed into an archive summary block appended to the system prompt.
11. AI Response Generation	Claude API called with assembled system prompt (guardrails, KB context, conversation stage, contact memory, language directive). Confidence score extracted from CONFIDENCE:0.XX marker.
12. Marker Extraction & Stripping	[SALES_LEAD], [STAGE:x], and [ENTITY:key=value] markers parsed and stripped before the message is sent to the customer.
13. Guardrail Enforcement	Blocked keywords/topics checked across all 4 layers. Response truncated to the effective max_response_length (most restrictive layer wins).
14. Escalation Policy	Low-confidence counter incremented. If consecutive low-confidence turns >= max_low_confidence_reprompts, conversation is auto-escalated.
15. Reply Dispatch	Message sent via provider gateway and stored with delivery_status=sent. Meta delivery/read receipts advance the status ladder: sent → delivered → read.
16. Lock Release	Processing lock cleared non-blocking. Conversation updated_at timestamp refreshed.

4.2 Conversation Management

Conversation Statuses

Status	Description
open	Bot is actively responding to the customer.
escalated	Conversation handed to a human agent; bot is silent.
bot_paused	Agent has taken manual control; messages are stored but not auto-replied.
resolved	Closed by an agent or auto-resolved after inactivity.

AI Conversation State Machine

Conversations progress through structured stages. The AI declares stage transitions and captures customer entities by appending control markers at the end of its response (stripped before delivery to the customer):

- greeting
- qualifying
- resolving
- following_up
- closing

Stage update: [STAGE:qualifying]

Entity capture: [ENTITY:email=alice@example.com] [ENTITY:order_id=ORD-123]

Delivery Status Ladder

Outbound messages move one-way: sent → delivered → read. The failed status is always accepted. Backward transitions are silently ignored.

Each tenant manages one or more KB Collections assigned to specific bots with a priority order. Documents are ingested, semantically chunked, and embedded via Voyage-3 into a pgvector index.

Supported Document Formats

PDF, DOCX, TXT, Markdown, Images (OCR via image extraction)

KB Entry Lifecycle

draft → review → live → archived

4-Strategy RAG Fallback

- Redis cache — previously retrieved results cached per (tenant, product, query).
- Vector similarity search — pgvector cosine similarity using Voyage-3 embeddings.
- Keyword ILIKE search — full-text keyword fallback when no vector match is found.
- Product-scoped legacy fallback — searches flat (non-collection) KB entries.

KB-Only Mode

When enabled at any guardrail layer, the AI answers only from KB results. If no relevant entry is found it acknowledges the gap and offers to escalate. Configurable globally, per product type, per tenant, or per individual bot.

AI-Generated KB Suggestions

Frequently asked questions not covered by the KB are flagged as suggestions (status: pending). Client managers review and promote them to live entries or dismiss them.

4.4 Escalation Management

Escalation Triggers

- Keyword match — customer types a configured phrase (e.g., 'speak to human', 'urgent', 'refund').
- Low-confidence consecutive turns — AI confidence below threshold for N turns in a row.
- Sales lead detected — [SALES_LEAD] marker triggers immediate handoff to a sales agent.
- KB-only no-match — KB-only mode active but no relevant KB entry was found.
- Auto-escalation timeout — conversation open for more than X configured hours.

Escalation Flow

- Conversation status flipped to 'escalated'; bot goes silent.
- Escalation record created (status: pending).
- Email notification dispatched to client_manager and platform_admin via Brevo.
- If auto_dispatch_on_escalation is enabled: outbound voice call placed to the customer.
- Agent claims the escalation → conversation assigned (agent_id set on conversation).
- Agent sends manual WhatsApp replies until they mark the escalation resolved.
- On resolution: conversation closed; optional CSAT prompt sent to the customer.

Human Takeover (bot_paused)

An agent can pause the bot without a full escalation — useful for brief interventions. Inbound messages are stored but not auto-replied. The agent can resume the bot at any time by switching status back to 'open'.

CSAT Collection

After resolution the bot sends a 1–5 star rating prompt. The score is stored in contact.memory_json with a timestamp. A confirmation message is sent (e.g., "**** Thank you for your feedback!").

4.5 Voice Call Module

The voice module enables AI-powered outbound phone calls via a configurable three-component stack: Telephony + Speech-to-Text + Text-to-Speech. Each call runs a multi-turn AI conversation loop with full transcript storage and automated structured outcome extraction.

Supported Voice Providers

Component	Providers	Notes
Telephony	Twilio, Exotel	Exotel recommended in India (lower per-minute cost). Twilio for international.
STT	Deepgram Nova-2, Sarvam AI, Azure Speech	Sarvam recommended for Hindi and Indian regional languages.
TTS	Twilio Say (free), Google Neural2, Sarvam TTS, Exotel Say	Twilio Say is zero-cost built-in; Google/Sarvam for higher audio quality.

Voice Call Flow

- Dispatch request (API, campaign trigger, or escalation auto-dispatch) creates a voice_calls record.
- Telephony provider places the outbound call.
- On answer: TwiML/ExoML greeting message played; customer prompted to speak.
- Customer audio → STT transcription → Claude AI response → TTS audio returned as next prompt.
- Multi-turn loop repeats until max_turns reached, silence timeout, or customer hangs up.
- Final status received from provider: completed / failed / voicemail / no_answer / busy.
- Claude analyses the full transcript and extracts structured outcome JSON.
- Call cost in INR calculated and stored alongside the full transcript.

Post-Call Outcome Extraction

Claude analyses the transcript and returns structured fields:

- intent
- product_interest
- resolved (boolean)
- escalation_needed (boolean)
- sentiment
- follow_up_action
- summary

Voice Configuration Options (per-bot)

- telephony_provider, stt_provider, tts_provider — override platform-level defaults
- language — en-IN, hi-IN, ta-IN, te-IN, kn-IN, mr-IN, gu-IN, ml-IN, en-US
- tts_voice — provider-specific voice name (e.g., Polly.Aditi, en-IN-Neural2-A)
- max_call_duration_seconds — maximum call length (default: 300 s)
- max_turns — maximum AI conversation turns per call (default: 20)
- silence_timeout_seconds — silence before hang-up (default: 5 s)
- voicemail_enabled / voicemail_message — leave a voicemail on no-answer
- greeting_message — opening message (overrides the product default)
- auto_dispatch_on_escalation — auto-call customer when their conversation is escalated
- escalation_voice_delay_seconds — delay before dispatching the escalation call

Provider Cost Estimation

A cost-estimate API endpoint returns the blended cost per minute (INR) for any telephony + STT + TTS combination, enabling platform managers to compare configurations.

4.6 Campaign Engine

Campaigns enable bulk outbound engagement via WhatsApp, Voice, or both simultaneously. Per-contact delivery and response status is tracked with configurable retry logic.

Campaign Channels

Channel	Behaviour
WhatsApp	Sends a WhatsApp template message to all contacts. Tracks: sent, replied, failed per contact.
Voice	Dispatches outbound AI calls to all contacts. Tracks: initiated, answered, voicemail, failed.

draft → active (on launch) → paused (manual) → completed / cancelled

Retry Logic

Per-campaign `retry_config`: `retry_after_hours` (wait before retry) and `max_retries` (maximum attempts per contact). Contacts in failed state are retried within the window.

Real-Time Campaign Stats

`stats_json` tracks: total contacts, WhatsApp sent/replied/failed, voice answered/voicemail/failed — all displayed on the campaign detail page.

4.7 Follow-Up Automation

Proactively re-engages contacts whose conversation has been idle for a configurable number of days.

Configuration Options

- `enabled` — toggle on/off per bot type
- `idle_days` — days of inactivity before follow-up triggers
- `message_template` — the follow-up message text
- `max_follow_ups` — maximum follow-ups per conversation before stopping
- `contact_scope` — all contacts or a specific segment

Follow-Up Statuses

scheduled → sent / failed / cancelled

4.8 Orders & Payment Collection

Agents or the lifecycle bot can create orders within a conversation. Payment links (PhonePe / Razorpay) are generated and sent via WhatsApp. Confirmations arrive via webhook.

Order Workflow

- Create order: add line items (name, quantity, unit price) and set total.
- Select payment provider and generate a payment link.
- Payment link sent to customer via WhatsApp.
- Customer pays → provider webhook received at `/api/payments/{provider}/webhook`.
- Payment status updated to 'paid'; order status updated to 'confirmed'.
- Conversation continues — bot can confirm delivery and handle post-payment queries.

Payment Providers

- PhonePe:** UPI-based payments for India. SHA256-signed payloads. Redirect to PhonePe checkout.
- Razorpay:** Multi-method: UPI, cards, net banking. Order API + webhook-driven confirmation.

Order Statuses

pending → confirmed → dispatched → delivered → cancelled

5. Configuration System

5.1 4-Layer Guardrail Cascade

Guardrails control what the AI can say and how it handles sensitive topics. Four layers are merged at runtime using these merge rules:

- Lists (`blocked_topics`, `blocked_keywords`): UNION — all layers contribute their lists.
- Numeric limits (`max_response_length`): MIN — most restrictive layer wins.
- Booleans (`kb_only_mode`, `no_external_links`, `no_personal_data`): OR — any layer can enable.
- Action/message strings (`on_blocked_topic`, `custom_blocked_message`): most specific layer wins.

#	Layer	Set By	Controls
1	Global Platform Settings	Platform Manager	<code>global_blocked_topics</code> , <code>global_blocked_keywords</code> ,

max_response_length (default 2000), enforce_kb_only_globally,
no_personal_data, no_external_links

Per product slug (support_bot / sales_bot / lifecycle_bot):
blocked_topics, blocked_keywords, max_response_length,
kb_only_mode, no_personal_data, no_external_links, on_blocked_topic

Per-bot overrides: tone, no_phone_numbers_in_response, on_low_confidence, reprompt_message, custom_blocked_message, max_response_length (default 1000)

Core Settings

- `system_prompt` — AI persona and business-specific instructions (overrides product default)
- `ai_model` — Anthropic model ID (e.g., `claude-sonnet-4-6`); overrides product default
- `confidence_threshold` — float 0–1; below this triggers low-confidence reprompt (default 0.6)
- `escalation_triggers` — keyword phrases for immediate human escalation
- `kb_only_mode` — restrict AI to KB answers only

Escalation Policy

- `confidence_threshold` — per-bot override of the platform default
- `max_low_confidence_reprompts` — consecutive low-confidence turns before escalating (default 2)
- `on_exhaust` — 'escalate' or 'silent' when reprompt limit is reached
- `reprompt_message` — optional message sent to the customer before the escalation turn
- `auto_escalate_after_hours` — auto-escalate after N hours open without resolution (null = off)

Tone & Content Filters

- `tone` — professional | casual | empathetic | formal (injected into system prompt)
- `no_external_links` — prevent the AI from including URLs
- `no_phone_numbers_in_response` — prevent the AI from sharing phone numbers
- `no_personal_data` — prevent the AI from referencing PII
- `on_blocked_topic` — escalate | silent | custom_message
- `custom_blocked_message` — text shown when a blocked topic is detected

5.3 LLM Configuration Hierarchy (6 Levels)

The platform resolves which AI model and API credentials to use through a 6-level hierarchy. The first valid match from the top wins:

Level	Source
1 — Client + Bot-specific	<code>llm_configs</code> row with matching <code>tenant_id</code> AND <code>product_slug</code>
2 — Client generic	<code>llm_configs</code> row with matching <code>tenant_id</code> , no <code>product_slug</code>
3 — Platform + Bot-specific	<code>llm_configs</code> row with no <code>tenant_id</code> , matching <code>product_slug</code>
4 — Platform generic	<code>llm_configs</code> row with no <code>tenant_id</code> , no <code>product_slug</code>
5 — Bot config model	<code>bot_configs.ai_model</code> (model only; uses platform API key)
6 — Product default	<code>products.default_model</code> (product catalog; uses platform API key)

Auth-failure fallback: if a client's custom API key returns a 401/403 error, the platform automatically retries the AI call using its own platform key before returning an error to the customer.

5.4 WhatsApp Number Management

Each tenant can configure one or more WhatsApp Business numbers, each scoped to a specific bot product and provider.

Provider	Configuration Required
Meta Cloud API	<code>phone_number_id</code> , <code>access_token</code> , <code>verify_token</code> , <code>app_secret</code> — official WhatsApp Business API.
Twilio	Account SID, Auth Token, WhatsApp-enabled phone number.
Interakt	API key and endpoint — third-party BSP.
WATI	API token and endpoint — third-party BSP.
Gupshup	API key and source phone — third-party BSP.

Numbers can be labelled, toggled active/inactive, and updated without service interruption. Multiple numbers per tenant are supported.

6.1 Client Dashboard Analytics (/analytics)

Available with 7-day and 30-day toggles.

- Total, open, escalated, resolved, and bot_paused conversation counts
- Daily message volume trend (bar chart, Recharts)
- Monthly AI token consumption rolling total
- Escalation rate and average resolution time
- Per-product breakdown (support_bot, sales_bot, lifecycle_bot)
- Contact sentiment distribution (positive, neutral, negative, frustrated)

6.2 Platform-Level Analytics (/platform/analytics)

- Total active clients and active trials
- Platform-wide conversation volume across all tenants
- Trial utilisation rates and paid conversion trends
- Per-client breakdown of message and token usage

6.3 Usage Events (Metered Billing Data)

Every interaction generates a usage_events row for billing and capacity planning:

Event Type	Triggered When
conversation_started	A new conversation is created for a contact.
message_sent	Each inbound message or bot reply.
ai_token_used	An AI response is generated (input + output token count stored).
escalation	A conversation is escalated to a human agent.
kb_query	A KB lookup is performed for an incoming message.

7. Notification System

Event-driven email notifications sent via Brevo (Sendinblue). Rules are configurable per event at platform-wide or per-tenant scope with role-based or custom email targeting.

Notification Events

Event	Description
trial_expiring_7d	Trial expires in 7 days — alert client manager.
trial_expiring_1d	Trial expires tomorrow — alert client manager + platform manager.
trial_expired	Trial ended — notify both parties.
client_invited	New client invite sent via the platform.
client_activated	Client completed onboarding and went live.
escalation_created	New escalation — notify assigned agent and managers.
escalation_timeout	Escalation unclaimed past the threshold duration.
new_client_onboarded	Platform manager alert on new client signup.
daily_report	Daily digest: conversation count, escalation rate, token usage.
low_confidence_spike	Unusual spike in low-confidence AI responses detected.
bot_error	Runtime error in the bot pipeline.
subscription_renewed	Billing cycle renewed successfully.

Recipient Targeting

- Role-based: platform_admin, platform_manager, client_manager, client_admin

Cost per platform-wide (all tenants) or per tenant (enterprise client only)

8. Free Trial Management

Platform managers can grant time-limited free trials per product per client. Trials operate independently from paid subscriptions.

- Duration: configurable per trial (default 14 days).
- Scope: per product — clients can trial each bot type independently.
- Model restriction: trial may be limited to a specific AI model.
- Status flow: active → expired or active → converted (on subscription).
- Automatic notifications: 7-day warning, 1-day warning, and expiry notification.
- Conversion tracking: flagged when a trial converts to a paid subscription.
- Trial utilisation tracked in platform analytics for conversion rate reporting.

9. Team Management

9.1 Client Team

Client managers invite team members via email. Invite links expire in 7 days.

Role	Permissions
client_manager	Full workspace control: bot configs, guardrails, KB, team, WhatsApp numbers, LLM keys, orders, billing.
client_admin	Escalation management, limited team management within the tenant.
agent	View/claim assigned escalations; send WhatsApp replies during human takeover.

9.2 Platform Team

Platform managers invite internal SaaS operator staff.

Role	Permissions
manager	Full platform control: create and manage all clients, products, configs.
admin	Read-only monitoring access to all client data for support and investigation.

10. Third-Party Integrations

Service	Category	Usage
Meta Cloud API	WhatsApp	Primary WhatsApp Business API. Webhook verification, message send, delivery receipts.
Twilio	WhatsApp + Voice	WhatsApp Sandbox + outbound/inbound AI voice calls (Twiml).
Interakt / WATI / Gupshup	WhatsApp BSP	Alternative third-party WhatsApp providers.
Exotel	Telephony (India)	India-focused outbound voice (ExoML). Lower per-minute cost than Twilio.
Deepgram Nova-2	STT	Fast, accurate English speech-to-text transcription.
Sarvam AI	STT + TTS	Indian regional language support: hi-IN, ta-IN, te-IN, kn-IN, mr-IN, gu-IN, ml-IN.
Azure Speech	STT	Enterprise-grade STT (optional).
Google Neural2 TTS	TTS	High-quality TTS for English and Indian regional languages.

Identity & Access

- tenants — Workspace/organisation (plan, status, provider).
- tenant_users — Team members linked to a tenant (role, user_id).
- platform_users — Platform staff (role: manager | admin).
- client_invites — Invite tokens (email, role, expires in 7 days).

Communication

- whatsapp_numbers — Business numbers (provider, product_slug, config_json with credentials, active flag).
- contacts — Customer records (phone/BSUID, name, memory_json: preferences, sentiment, CSAT, order history).
- conversations — Instances (status, product_type, stage, ai_vars, assigned_agent_id, processing_lock).
- messages — History (role, content, media_url, confidence_score, delivery_status, whatsapp_msg_id for dedup).
- escalations — Records (trigger_reason, agent_id, status: pending | assigned | resolved).
- agent_sessions — Human takeover periods (started_at, ended_at, resolution_note).

AI & Configuration

- products — Product catalog (slug, name, default_prompt, default_model, active).
- bot_configs — Per-bot settings (system_prompt, confidence_threshold, guardrails_json, voice_config, escalation_policy).
- platform_settings — Global key-value store for guardrails and feature flags.
- bot_type_guardrails — Layer 2 guardrails scoped to a product_slug.
- tenant_guardrails — Layer 3 guardrails shared across all bots for a tenant.
- llm_configs — LLM credentials (tenant_id x product_slug hierarchy; api_key, model, validation_status).

Knowledge Base

- kb_collections — Named KB collections (tenant, name, description).
- kb_collection_bots — Assigns collections to bots with priority ordering.
- knowledge_base — KB entries (question, answer, category, embedding vector, status, version).
- kb_documents — Uploaded documents (PDF/DOCX/TXT; status: pending → done | error; chunk_count).
- kb_suggestions — AI-generated FAQ suggestions (frequency count, status: pending | accepted | dismissed).

Commerce

- orders — Customer orders (items_json, total, status, contact_id, conversation_id).
- payments — Payment records (order_id, provider, status, link_url, webhook_received_at, payment_ref).
- campaigns — Outbound campaigns (channel, template_id, status, stats_json, retry_config).
- campaign_contacts — Per-contact campaign tracking (whatsapp_status, voice_status, attempts).
- follow_up_configs — Follow-up settings (idle_days, max_follow_ups, message_template).
- follow_up_sequences — Scheduled follow-up send log.

Voice & Telephony

- voice_provider_configs — Provider registry (component: telephony | stt | tts, credentials, cost_per_min_inr).
- voice_calls — Call log (status, duration_seconds, turn_count, transcript, outcome_json, cost_rupees, triggered_by).

Billing & Usage

- free_trials — Time-limited trials (tenant_id, product_slug, starts_at, ends_at, status).
- subscriptions — Billing records (product, tier, billing_cycle, next_billing_date).
- usage_events — Metered usage rows (event_type, token_count, timestamp).
- notification_configs — Event-triggered email rules (scope, event_type, recipients, enabled).
- tenant_products — Product activation per tenant (tier: base | advanced).

12. Security & Data Isolation

12.1 Row-Level Security (RLS)

Supabase RLS policies enforce strict per-tenant data isolation at the database level. All web app queries run under

- Tenants can only read/write their own conversations, contacts, messages, and KB.
- Agents see only conversations/escalations belonging to their tenant.
- Platform staff: admin (read-all) vs manager (read-write-all) enforced by policy.
- `get_user_tenant_id()` Postgres function resolves the caller's tenant from JWT claims.

12.2 Credential & Secret Management

- WhatsApp credentials (access tokens, phone number IDs) stored encrypted in `config_json` per number.
- Payment credentials (merchant IDs, salt keys) held in environment variables only.
- LLM API keys stored in `llm_configs`; masked in all API responses.
- Voice provider credentials in `voice_provider_configs.credentials_json`.
- Webhook signatures validated (Meta HMAC `app_secret`, Razorpay webhook secret).
- Environment variables managed via deployment platform — never committed to source control.

12.3 Anti-Abuse Controls

- 30-second optimistic lock prevents duplicate AI calls from Meta webhook retries.
- `whatsapp_msg_id` unique constraint prevents duplicate message storage.
- Plan limits (conversation count + monthly token quota) checked before every AI call.
- Suspended tenant check at webhook entry — no processing for suspended accounts.
- Blocked keyword/topic guardrails enforced at every layer before sending any AI response.

13. Non-Functional Requirements

13.1 Performance

- Webhook HTTP response < 100 ms (200 sent immediately; AI work is fully async).
- Bot context Redis cache (60 s TTL) reduces Supabase round-trips on hot paths.
- KB lookup target < 300 ms (Redis cache hit < 5 ms; vector search < 200 ms).
- `get_bot_context` RPC consolidates 7 Supabase queries into a single database call.

13.2 Reliability & Resilience

- Meta webhook retries handled via optimistic lock + `whatsapp_msg_id` deduplication.
- AI API key fallback: client key auth failure → automatic retry with platform key.
- Provider inference fallback: wrong WhatsApp provider inferred → retry with alternate.
- Graceful degradation: if AI pipeline fails, customer receives a human-escalation message.
- Voice call status webhooks handle all terminal states (completed, failed, voicemail, no_answer, busy).

13.3 Scalability

- Stateless API server — horizontal scaling supported.
- Redis caching for high-frequency bot context and KB lookup paths.
- pgvector for semantic KB search at large embedding volume.
- Metered usage events provide real-time billing and capacity planning data.

13.4 Internationalisation

- Auto language detection: Hindi (Devanagari), Tamil, Telugu, Kannada, Punjabi, Gujarati, Odia, Arabic, Chinese.
- Bot auto-replies in the detected language/script when non-Latin text is received.
- Voice STT/TTS supports 8+ Indian regional language codes (en-IN, hi-IN, ta-IN, te-IN, etc.).
- Payment providers (PhonePe, Razorpay) and telephony (Exotel) optimised for India.

14. Glossary

Term	Definition
BSUID	WhatsApp Business Unique Identifier — opaque ID used instead of a phone number for users with a WhatsApp

